

Workflow Documentation

SupportFlow AI

Autonomous AI-Powered Customer Support Automation System

1. Project Overview

Project Name

SupportFlow AI — Autonomous Customer Support Automation System

Objective

SupportFlow AI is an AI-powered customer support automation platform built using **n8n workflow automation, Groq LLMs, Google Sheets, and Gmail API.**

The system automates the complete customer support lifecycle:

- Customer ticket creation
- AI-based ticket analysis
- Priority and sentiment detection

- Automatic department routing
- Agent assignment
- AI-generated customer responses
- Resolution tracking
- Feedback collection

The objective is to reduce manual support workload, improve response time, and create an intelligent support pipeline capable of handling customer queries autonomously.

2. System Architecture Overview

High-Level Flow

Customer



Ticket Submission

(Google Forms / Email)

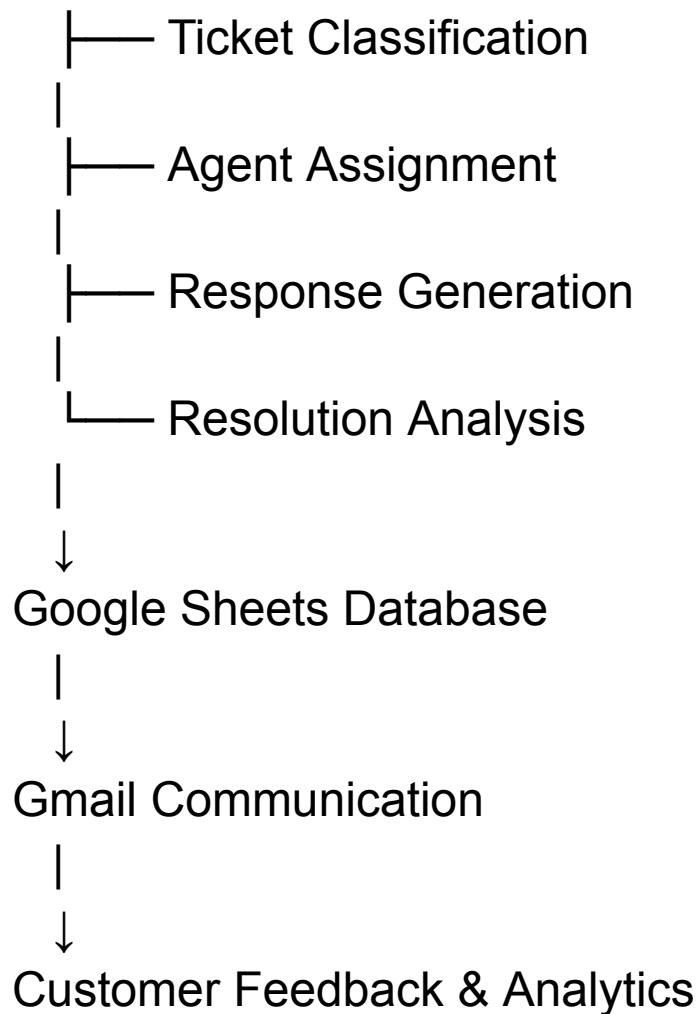


n8n Automation Platform



AI Processing Pipeline





3. Technology Stack

Component	Technology
Workflow Automation	n8n
AI Model Provider	Groq API

Large Language Model	Llama Models
Database	Google Sheets
Communication	Gmail API
User Input	Google Forms
Programming Logic	JavaScript Code Nodes

4. Database Structure

The project uses Google Sheets as a lightweight database.

4.1 Tickets Sheet

Stores all customer support requests.

Field	Description
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ticket_id	Unique ticket identifier
customer_name	Customer name
customer_email	Customer email
subject	Ticket subject
message	Customer issue
category	AI detected category
priority	AI priority level
sentiment	Customer sentiment
department	Assigned department
assigned_agent	Responsible agent
status	Current ticket status

ai_respo nse	Generated response
resolutio n_note	Resolution details
feedback _score	Customer rating

4.2 Agents Sheet

Stores support team information.

Field	Description
agent_ name	Support agent
depart ment	Assigned department
agent_ email	Contact email

Example:

Agent	Department
Vidhaan	Billing

4.3 Feedback Sheet

Stores customer feedback.

Field	Description
ticket_id	Ticket reference
rating	Customer rating
comments	Feedback message

WORKFLOW 1

Ticket Intake & Creation

Purpose

The first workflow collects customer queries and creates structured support tickets.

Workflow Diagram

Customer



Google Form



n8n Trigger



Google Sheets



Create Ticket Record

Nodes Explanation

Node 1: Google Form Trigger

Function

Receives customer support requests.

Input

Customer provides:

- Name
- Email
- Subject
- Problem description

Example:

Name:

Rahul

Issue:

Charged twice for payment

Node 2: Google Sheets — Add Ticket

Function

Stores customer information in the ticket database.

The node creates:

ticket_id

status = New

created_at

customer details

Output

Example:

```
{  
  "ticket_id":"TICKET-1786189064209",  
  "status":"New",  
  "message":"Charged twice"  
}
```

WORKFLOW 2

AI Ticket Classification

Purpose

Analyze customer issues using AI and extract important information.

Workflow Diagram

New Ticket

|

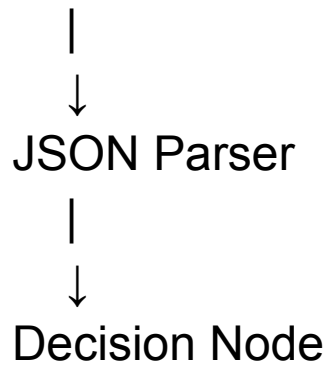
↓

Google Sheets Read

|

↓

Groq AI Agent



Nodes Explanation

Node 1: Google Sheets Read

Function

Fetches newly created tickets.

The workflow retrieves:

- Customer message
 - Subject
 - Ticket ID
-

Node 2: Groq AI Agent

Function

Uses LLM reasoning to understand the customer issue.

The AI analyzes:

Category

Examples:

- Billing
- Technical
- Account
- Shipping

Priority

Examples:

- High
- Medium
- Low

Sentiment

Examples:

- Positive
- Neutral
- Negative

Department

Routes ticket to correct team.

Example AI Output

```
{  
  "category": "Billing",  
  "priority": "High",  
  "sentiment": "Negative",  
  "department": "Billing",  
  "human_required": true,  
  "summary": "Duplicate payment issue"  
}
```

Node 3: Code Node

Purpose

Clean and format AI response.

The AI output may contain:

```
```json  
{
 ...
}
```

The Code Node removes unwanted formatting and converts it into valid JSON.

---

# Node 4: IF Node

## Purpose

Checks whether human intervention is required.

Condition:

human\_required = true

---

## TRUE Path

Human Required

Ticket is escalated.

---

## FALSE Path

Continue Automation

Ticket moves to next workflow.

---

## # WORKFLOW 3

### # Automatic Agent Assignment

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#### ## Purpose

Automatically assign tickets to the correct support agent.

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#### ## Workflow Diagram

Classified Ticket

|

↓

Department Detection

|

↓

Google Sheets Lookup

|

↓

Assign Agent

|

↓

Update Ticket

---

# Nodes Explanation

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# Node 1: Department Extraction

Reads:

Department = Billing

---

# Node 2: Google Sheets Lookup

Searches Agents Sheet.

Example:



Input:

Billing

Output:

Agent:

Vidhaan

Email:

agent@gmail.com

---

# Node 3: Code Node

Formats assignment data.

Output:

```json

{

"department":"Billing",

"agent_name":"Vidhaan",

```
"agent_email":"agent@gmail.com"  
}
```

Node 4: Update Ticket

Updates database:

```
assigned_agent = Vidhaan  
status = Assigned
```

WORKFLOW 4

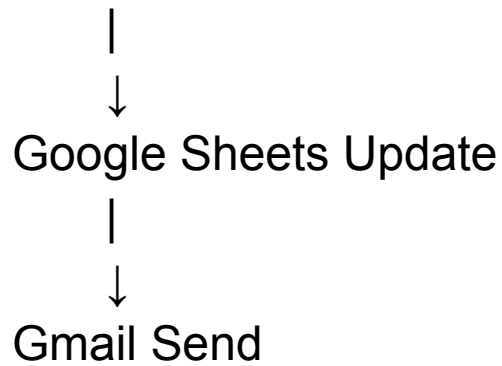
AI Customer Response Generation

Purpose

Generate personalized responses automatically.

Workflow Diagram

```
Assigned Ticket  
  |  
  ↓  
Groq AI Agent
```



Node Explanation

Node 1: Retrieve Assigned Ticket

Gets:

- Customer details
 - Issue
 - Classification
 - Assigned agent
-

Node 2: AI Response Agent

Generates professional customer replies.

Prompt objective:

- Be polite

- Explain issue status
 - Provide next steps
 - Maintain professional tone
-

Example:

Input:

Charged twice for payment

Output:

Hello Rahul,

We have received your billing complaint.
Our team is reviewing the duplicate payment issue.
You will receive an update soon.

Node 3: Google Sheets Update

Stores:

ai_response

status = Response Sent

Node 4: Gmail

Sends response automatically.

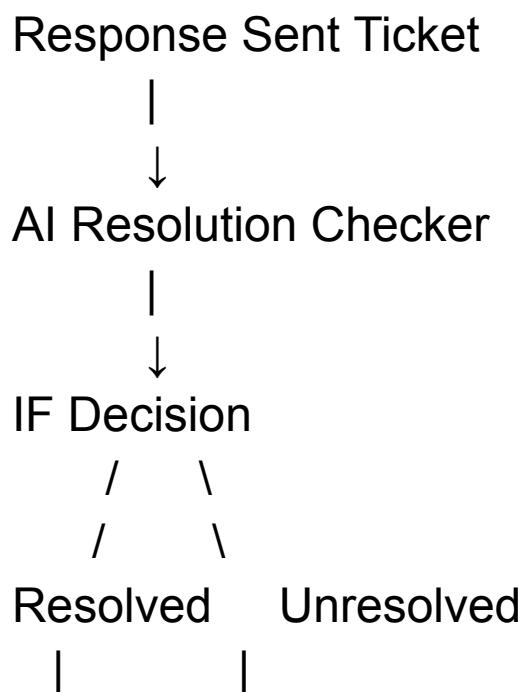
WORKFLOW 5

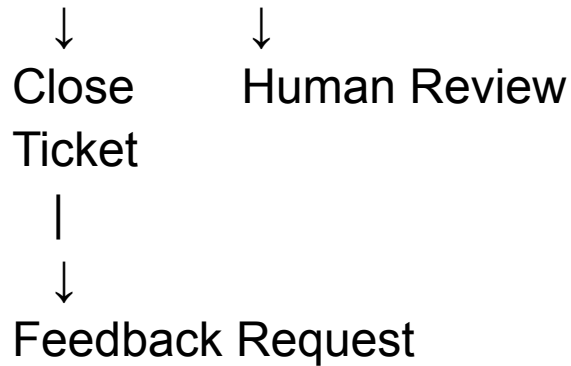
Resolution, Closure & Feedback

Purpose

Determine whether tickets are solved and complete the support cycle.

Workflow Diagram





Node Explanation

Node 1: Schedule Trigger

Runs periodically and checks pending tickets.

Node 2: AI Resolution Checker

Determines:

resolved = true/false

Example:

```
{  
  "resolved":true,  
  "reason":"Customer issue solved"  
}
```

Node 3: Code Node

Converts AI output into structured JSON.

Node 4: IF Node

Decision:

TRUE

Ticket is resolved.

Actions:

- Close ticket
- Add closing timestamp
- Send feedback request

FALSE

Issue requires attention.

Actions:

- Mark Human Required
 - Notify support team
-

Node 5: Google Sheets Update

Updates:

status = Closed

closed_at = timestamp

resolution_note

Node 6: Gmail Feedback Request

Sends:

Your ticket has been resolved.

Please rate your experience from 1-5.

Node 7: Feedback Storage

Stores:

- Rating
 - Comments
 - Ticket reference
-

5. Complete System Execution Example

Customer Input

Subject:

Payment Issue

Message:

I was charged twice

AI Classification

Category:

Billing

Priority:

High

Sentiment:

Negative

Department:

Billing

Agent Assignment

Agent:

Vidhaan

Team:
Billing

AI Response

Your billing issue has been received.
Our team is investigating the duplicate payment.

Resolution

Status:
Closed

Feedback

Rating:
5/5

6. Error Handling Strategy

AI Failure

If Groq API fails:

- Retry request
- Log error

- Escalate to human
-

Database Failure

If Google Sheets fails:

- Stop workflow
 - Preserve ticket information
 - Notify administrator
-

Email Failure

If Gmail fails:

- Retry sending
 - Store unsent response
-

7. Future Enhancements

RAG Knowledge Base

Connect company documentation so AI can answer from internal knowledge.

Multi-Agent AI Architecture

Separate agents:

- Classification Agent
 - Routing Agent
 - Response Agent
 - Quality Control Agent
-

Analytics Dashboard

Track:

- Ticket volume
 - Resolution time
 - Customer satisfaction
 - Agent performance
-

8. Conclusion

SupportFlow AI demonstrates how AI agents and workflow automation can transform traditional customer support into an intelligent autonomous system.

By combining **n8n automation**, **LLM-based reasoning**, **structured databases**, and **communication APIs**, the system provides faster responses, intelligent routing,

reduced manual effort, and scalable customer service automation.

Project: SupportFlow AI

Category: AI Automation / Agentic AI / Customer Support

Built Using: n8n + Groq LLM + Google Sheets + Gmail API